

Finding optimal solutions to a Teacher Assignment Problem using MILP- and SMT-Solvers

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Abstract

The area of Teacher Assignment Problems is a topic of high interest, not only for schools and universities. Therefore, over the years many different solution-strategies evolved, each trying to keep computation time low whilst still finding an optimum or near-optimum. We look at a specific real-world version of this problem: teachers have to be assigned to courses in geographically different locations according to their distance, preferences, as well as availability constraints. We reformulate the problem as a linear optimization problem over integer variables by linearizing the non-linear parts of the objective function. Two different approaches for the optimization are used and compared: Using the MILP-Solver Pulp and using the SMT-Solver Z3 with the vZ extension.

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1 Introduction

1.1 Overview over Teacher Assignment Problems

- what is a teacher assignment problem?
- Transportation Problem: suppliers with supply, destinations with demand, edge between all suppliers and destinations, cost for each edge, decision variable for each edge
- Assignment Problem: any source (agent) can be assigned to any destination (task), each edge has a cost and a binary decision variable. Demand of each source is 0 or 1.
- Teacher Assignment Problem: TP, but demand can be bigger than 1, not every source is connected to every destination, additional objective regarding distribution of workdays and additional constraints
- origin of teacher assignment problems / typical use cases

1.2 Related work

- overview of scientific material, different strategies, algorithms

1.3 Problem description

- describing the concrete problem which is being solved in this thesis
- stress that also unsatisfiable scenarios should be considered

2 Formulating the problem

- Linear Solvers as flexible, fast way to find perfect optimum → aiming to find linear cost function and linear constraints

2.1 Cost function

- what needs to be considered?
- how are the different costs balanced? normalization, possibility of custom weights
- problem of standard deviation (not linear) → optimization of average relative deviation and maximum relative deviation
- formula of the cost function

2.2 Constraints

- formulas of the constraints
- explaining the workaround to get average relative deviations, as well as the maximum of them

2.3 Balancing the weights

- How is standard deviation best approximated (in my use case)
- How should you weigh standard deviation against priority against distance

3 Computing the optimum

- ILP Problem, therefore using the MILP Solver pulp makes sense
- vZ is tried, to see how it compares

3.1 Comparison of the solvers

- What goes on inside of PuLP?
- PuLP easy to handle, no issues
- What goes on inside of vZ?
- vZ has issues of mixing Ints and Reals in the objective function
- a custom binary search was implemented for Z3 in order to create custom cutoff-point

3.2 Considering unsatisfiability

- use Z3 to find out which teachers have to be available additionally for which events, in order to make it satisfiable
- the problem is solved with less strict constraints regarding the needed number of teachers for these events

4 Evaluation

4.1 Execution Time

- comparison of the different execution times - only checking satisfiability with Z3 and optimizing with PuLP, vZ and the binary implementation (with various cut-off points)

- conclusion: PuLP much faster than vZ and also for Z3 (if using realistic cutoff points), checking for satisfiability very fast
- → PuLP is used in the final version of the program

4.2 Output Quality

- the calculated solutions are sensible and in some ways better than the solutions obtained by hand
- it becomes apparent that lots of bias is included when assignment process is done by hand - intended or unintended
- many more soft objectives have to be included, e.g. teaching quality of teachers, to take all of the intended bias into account
- making it possible for the end-user to pre-enter teachers for certain events and to change weights, makes it possible to find a good solution in little time, even with very specific objectives/constraints

5 Conclusion

- is the problem solved?
- what are the advantages / disadvantages of this approach
- which solvers proved to be useful